WEIGHTED FEDERATED SYNTHETIC DATA GENERATION

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by

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CERTIFICATE

This is to certify that the work contained in this project report entitled "Weighted Federated Synthetic Data Generation" submitted by Shivansh Gupta (Roll No.: 230123058) to the Department of Mathematics, Indian Institute of Technology Guwahati towards partial requirement of Bachelor of Technology in Mathematics and Computing has been carried out by him under my supervision.

It is also certified that, along with literature survey, computational implementations, simulation studies and empirical analysis have been carried out by the student under the project.

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ABSTRACT

Five offices hold records with the same columns but different populations. If each office supplies a fitted generator, how often should the server sample from each one? Choosing in proportion to training-set size reproduces the intended client mix. Choosing weights to match a few reported statistics might improve accuracy, but it can also change whose records the synthetic table represents. This project measures that trade-off for non-neural generators on UCI Adult and tests the conclusion on Bank Marketing.

The comparison fixes the real reference before estimating weights and measures the largest difference between query answers on real and generated tables. At the registered margin of 0.005, the constrained statistical rule did not improve on size weighting in any of 18 Adult development slices, six frozen Adult slices or four Bank slices. A different intervention had a much larger effect: enforcing Adult's documented education-code relation on the same generated rows reduced the Chow–Liu maximum error from 0.268–0.328 to 0.028–0.033 in the controlled comparisons.

The contribution is a way to separate errors that weighting can correct from errors in the local generators. A proof gives conditions under which size weights are optimal; a two-client example shows why the conclusion fails for imperfect models. Six studies connect those limits to finite sampling, conditional requests, prediction, schema constraints and the cost of release. The main generator does not carry a differential privacy guarantee.

Reproducibility: GitHub repository: S-G-mathematics/btp-federated-synthetic-data.

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Chapter 1

Data and Research Question

An analyst wants a synthetic table describing several populations held at separate sites. The sites share columns but serve different people. Giving each site equal weight may misrepresent the combined population. When local generators are imperfect, changing their sampling weights can correct aggregate errors or suppress a difficult-to-model site.

We study this problem with the UCI Adult dataset [1]. Each record contains 14 predictors—six numerical and eight categorical—and a binary income label. For example, the proportion of records with high-school graduation and income above $50,000 depends on a feature–label relationship, not just the separate frequencies. These clients are simulated, not participating institutions.

Table 1.1: The Adult schema. Every simulated client keeps the same columns.

Type	Recorded variables
Numerical (6)	Age, final weight, education number, capital gain, capital loss and hours worked per week.
Categorical (8)	Work class, education, marital status, occupation, relationship, race, sex and native country.
Binary outcome	Whether annual income exceeds $50,000.

The experiments create five or ten clients with a common schema and non-identical distributions. Assignment depends on age, age and education together, or the income label. The training-fitted assignment rule is reused across data roles. Label-skew assignment uses held-out labels to construct the benchmark and is not a service for routing people with unknown labels.

Each client fits and tunes a joint model of its features and label. The server receives fitted models, declared metadata and specified aggregate calibration summaries, then samples synthetic records without receiving original rows through that interface. A separate evaluator uses held-out real records. The main pipeline is nonprivate: withholding rows does not establish differential privacy [4].

1.1 Data roles and replication

Adult supplies 32,561 official training records and 16,281 official test records. Twenty-five training rows exactly matching complete test records are quarantined; remaining duplicate groups stay together across development splits. The usable training data provide local training, tuning, calibration and validation roles, with validation further separated for selection and development assessment. `fnlwgt` is a predictor, not a survey weight.

The official test is the frozen confirmation reference. It was read earlier for source validation and duplicate quarantine, but first scored for confirmation after the procedure was fixed. Development informed preprocessing and the workload; confirmation configurations reuse the same test people.

The external replication uses the 41,188 records in the *additional-full* version of UCI Bank Marketing, with a term-deposit subscription label [9]. Post-contact duration is removed before modelling. A duplicate-group-aware random split supplies training, tuning, calibration, validation and assessment roles; it is not a chronological split, and contacts may repeat people. The replication transfers two generator families to five-client age-and-education and label-skew assignments.

1.2 Question and contribution

The intended target weights clients by their training-record counts. Size weighting uses those same proportions to sample local generators. A constrained alternative uses calibration summaries and synthetic pilot answers to adjust the weights while limiting departure from the target and retaining at least half of each client's target mass. The

held-out reference keeps the original target proportions, allowing error correction to be assessed without changing the represented population.

The question is whether constrained statistical weighting materially improves empirical fidelity over a properly tuned size-weighted mixture under this information contract. The primary Adult metric is the largest real–synthetic difference across 152 bounded queries: 143 one-way frequencies, eight two-way queries and one education-code check. Conditional generation and prediction are assessed separately: finite-query accuracy does not guarantee either.

Calibration and weighting are established ideas: entropy balancing matches selected moments [5], kernel mean matching addresses sample selection bias [6], and predictive stacking optimizes a predictive objective [10]. Federated synthesis and heterogeneity also precede this work, including FLAIM's measurement-based approach and Fed-TGAN's similarity-aware weighting [7, 11]. This study's contribution is a controlled comparison of fixed, matched client generators against a fixed target, with representation constraints, frozen confirmation and external replication; it claims neither a new weighting principle nor a new generator architecture.

Chapter 2

Mathematical Framework

A complete record is $Z = (X, Y)$ on a common product schema: X contains mixed numerical and categorical attributes and Y is the binary outcome. A client partition changes which rows are observed, while preserving the columns. Let D_i^{tr} be client i's training table, with n_i records, and let $\alpha_i = n_i / \sum_j n_j$. The declared size-standardized target gives client i mass α_i; it is not itself a choice of generator weights. Write P_i for a hypothetical client population law and Q_i for its fitted and released generator. A server-side mixture uses $w \in \Delta_K = \{w \geq 0 : \sum_i w_i = 1\}$ and samples from $Q_w = \sum_i w_i Q_i$. Unless $w = \alpha$, latent client contributions differ from the declared target proportions even when some released query answers happen to match.

The information available to the server matters. Fitted generators can provide model samples and, where supported, model probabilities. They do not by themselves reveal the real reference answers or the local fitting errors. Any real-data calibration summaries used for selecting weights are additional releases, subject to their own access and privacy assumptions. A validation reference or held-out real reference is a separate information source. In particular, a method selected using such a reference must not be described as model-only.

2.1 Finite reference and query objective

Fix a finite workload $\mathcal{Q} = \{q_1, \ldots, q_L\}$ of measurable functions with values in $[0, 1]$. These may encode marginal, joint, or other bounded statistics; the particular workload

determines what is evaluated. Let D_i^R be the real reference records for client i, and define

$$B_{\ell i} = \frac{1}{|D_i^R|} \sum_{z \in D_i^R} q_\ell(z), \qquad b_R = B\alpha, \qquad H_{\ell i} = \mathbb{E}_{Q_i} q_\ell. \tag{2.1}$$

Thus b_R is the α-standardized real reference vector, even if reference sample sizes differ from the original n_i. If a client has positive target weight but no reference rows, the endpoint is unsupported. For a generated table D_S of m rows, b_S is its vector of ordinary query averages. The realized finite-table objective is

$$\mathcal{L}_{\mathcal{Q}}(D_S, D_R; \alpha) = \|b_S - B\alpha\|_\infty. \tag{2.2}$$

It is a maximum over this finite workload, not a claim about all queries or population loss. The analytical model-expectation counterpart is $L_R(w) = \|Hw - b_R\|_\infty$. A finite synthetic draw and an exact model expectation are different objects.

For stratified generation, allocate integer counts m_i with $\sum_i m_i = m$, independently draw m_i rows from Q_i, and pool or shuffle the rows. Put $\bar{w}_i = m_i/m$ and $\rho_m(w) = \frac{1}{2}\|\bar{w} - w\|_1$. Conditional on fixed models and selected weights, Hoeffding's inequality and a union bound give, with probability at least $1 - \gamma$,

$$\|b_S - Hw\|_\infty \le \sqrt{\frac{\log(2L/\gamma)}{2m}} + \rho_m(w). \tag{2.3}$$

Indeed, the expected table vector is $H\bar{w}$, each query average consists of independent bounded terms, and the allocation discrepancy is at most $\rho_m(w)$. Largest-remainder allocation gives $\rho_m(w) < K/(2m)$. Shuffling changes no query average. Independent mixture sampling has expectation Hw and the same concentration term without deterministic allocation error.

2.2 Feasible weights and operational selection

The retained representation constraints are

$$\mathcal{W} = \{w \in \Delta_K : \|w - \alpha\|_1 \le 0.25, \quad w_i \ge 0.5\alpha_i \text{ for all } i\}. \tag{2.4}$$

They cap total moved mass at 0.125 and preserve a positive floor for each client with $\alpha_i > 0$. They do not ensure that a generated row has any particular real-world attribute, nor that weighted query error improves. If $\alpha_i = 0$, the floor alone gives no positive representation guarantee.

A generic operational calibration objective uses an available pilot matrix $\widehat{H}$, a permitted summary $\widehat{b}$, and optional regularization:

$$\widehat{w} \in \arg\min_{w \in \mathcal{W}} \left\{ \|\widehat{H}w - \widehat{b}\|_\infty + \frac{\lambda}{2}\|w - \alpha\|_2^2 \right\}, \qquad \lambda \geq 0. \tag{2.5}$$

The retained constrained rule uses $\lambda = 0$. In the main Adult comparison its pilot has 1,500 rows per client and its real target estimate is the released calibration vector. Its linear program minimizes a finite maximum residual. The size baseline uses $w = \alpha$; other feasible challengers can differ in their source of weights or aggregation rule. A validation-selected challenger uses its validation answers, while the hindsight oracle minimizing L_R against the evaluation reference uses information unavailable at deployment. Neither should be conflated with a deployable model-only rule. If candidates also change the generator family or aggregation operator, changing weights alone does not describe the entire comparison.

Proposition 2.2.1 (Insufficiency of releases alone)**.** *Suppose two possible worlds release the same* $Q_1 = \text{Bernoulli}(1/4)$ *and* $Q_2 = \text{Bernoulli}(3/4)$*, with no restriction relating these models to the real reference. For the query* $q(z) = z$*, let the reference mean be* $1/4$ *in one world and* $3/4$ *in the other. Any release-only rule has worst-world expected absolute error at least* $1/4$*.*

Proof. The rule has the same output distribution in both worlds. For every output answer x, $|x - 1/4| + |x - 3/4| \geq 1/2$. Taking expectations shows that at least one world's expected error is at least $1/4$. The respective oracle weights are $(1, 0)$ and $(0, 1)$. □

This example concerns an unconstrained two-world identification problem; it does not estimate the experimental error. It shows why useful target summaries or assumptions linking models to real data are necessary for a general guarantee. Restricting weights or regularizing can select a unique answer without identifying the real target.

Chapter 3

From a weighting rule to a test

3.1 Fitting a client generator

Consider two clients with the same educational qualifications on their books but different age profiles. An age shift can be corrected by drawing more rows from the younger client. By contrast, if both generators assign the wrong numerical code to a qualification, transferring mass between clients need not fix that mistake. A comparison of weight rules must therefore start from the *same fitted and corrected local models*.

The development comparison used three non-neural families. The Gaussian copula fits marginal transforms and a dependence matrix in a latent Gaussian space. It can describe several pairwise associations at once, although the ordering assigned to a nominal variable affects the transform. A Chow–Liu tree fits a single tree of dependencies among discretized variables and samples from its smoothed conditional tables [2]. For integer attributes, sampling also specifies a value within the chosen bin. A mixed Gaussian mixture uses a latent component to couple the numerical and categorical coordinates. After transformation, rounding and clipping, its continuous density does not give the exact probability of emitting an integer row. These distinctions matter when a method asks a fitted generator for a probability rather than a sample.

All three emit rows through the same deterministic education-code correction. If the education category is recognized, the corresponding numerical code is set using the public mapping. An unmapped category remains visible to the structural diagnostic. The correction never consults a held-out real row. In the ablation, the same raw rows

can therefore be assessed on both sides of the transformation.

For the selected copula each client evaluated six settings: 8, 16 or 32 bins and smoothing 0.1 or 1. The fit used local training data. Among eligible fits, local tuning chose the smallest mean absolute primary-query error on 1,000 generated rows against local tuning records, with grid order resolving a tie. Eligibility included support, convergence, label validity and finite scores on the first 64 local tuning records. Only the choice among generator families used the separate global validation comparison.

3.2 What information each comparison receives

All rules can use client models and training counts. A rule that calibrates moments needs answers about the real clients as well. The retained challenger gets a vector of unnoised calibration query means, combined using the declared target shares. It estimates each generator's answers from a pilot sample and minimizes the largest absolute residual subject to the representation constraints. It need not receive the client's rows. Giving another rule a matrix of scores on individual real records, or showing it the separate validation answer vector, changes the question it can answer.

Table 3.1: Weighting and pooling candidates. A different information source or a different pooling operator changes the interpretation of a comparison.

Rule	Input beyond the model	Objective or role
Size; uniform	Counts for size weighting	Preserve the training shares; equal-client comparator.
Score and evidence rules; stacking	Scores for individual real calibration rows	Favor predictive scores or maximize mixture log score; privileged access.
Moment and kernel calibration	Real summary vectors and synthetic pilots	Match moments or finite feature means.
Robust and constrained calibration	Summaries, pilots and specified constraints	Control residuals; retained rule minimizes the largest query residual.
Geometric and transport pools	Finite candidate supports and scores or transport costs	Combine model masses by a different operator; finite searches only.
Validation LP	Separate validation answers and a fixed pilot	Privileged optimum of a finite program.

Predictive stacking [10], moment balancing [5], and transport barycenters [3] motivate different entries in Table 3.1. A mixture can score well on income prediction without matching the worst structural query. Likewise an optimizer can match its finite pilot well while its next synthetic draw performs less well. These objectives are related, but substituting one for another would overstate the comparison.

3.3 Freezing the comparison

Validation compared each candidate family using size weighting, five 10,000-row draws per cell and 18 development cells per family. Mean maximum query errors were 0.029975 for copula, 0.031455 for Chow–Liu and 0.051446 for the mixed Gaussian mixture. The registered 0.001 tie window did not change the decision. The projected copula with size weighting was selected, and the retained constrained rule was carried forward as its challenger. This was a global choice on validation data; earlier development diagnostics had already informed the study design.

The written freeze preceded scoring the official Adult test on seeds 404 and 505. Model settings, target, weighting, workload, draw size, support policy and downstream calibration were fixed at that point. Adult's test people recur in the different configurations. Five draws quantify generation variation conditional on a fit; neither those draws nor repeated partitions provide independent cohorts of people. For Bank Marketing the registered replication uses Bank-specific queries and an analogous comparison. Absolute loss levels across the two workloads need not be comparable.

There are three useful standards of evidence in what follows. An identity of fitted mixtures follows by algebra. A paired change on the same generated rows isolates the effect of a specified operation on those rows. A frozen comparison on held-out records tests a prediction about the procedure in a specified setting. The tables distinguish these roles because each warrants a different conclusion.

Chapter 4

Six experimental studies

The six studies address different questions about the same generator, from assigning client weights to releasing a usable table. Appendix A maps them to the archived work packages. A cell fixes a generator, client assignment, client count and fitting seed. Unless stated otherwise, five generated tables are assessed per cell. Primary loss is the maximum absolute discrepancy over the registered queries, with reference answers standardized to target client masses α. A constrained-minus-size difference $\Delta \leq -0.005$ was the practical-gain threshold.

Adult seeds reuse individuals and confirmation configurations share the official test people. Reported means and ranges are descriptive paired outcomes, not independent-cohort confidence intervals. Roles were simulated on one host; access restrictions describe interfaces, not a deployed network.

4.1 Study 1: Fixed-target weights and subset composition

Can weights chosen from released summaries improve the same fixed population target, while retaining client representation? The corrected development comparison crossed copula, Chow–Liu and mixed Gaussian mixture (GMM) generators with age, age-plus-education and label-skew assignments, $K \in \{5, 10\}$ clients, and seeds 101, 202 and 303. It therefore contains 54 fitting cells. The constrained rule used an ℓ_1 distance budget of

0.25 from the size weights and a floor of $0.5\alpha_i$ for each client. Each method generated five tables of 10,000 rows per cell. A slice averages the three seed-level cell means for one generator, partition and K combination.

The development mean was +0.002428; no slice met the margin (Table 4.1), although three individual cells did. An unconstrained validation-selected rule achieved −0.006951 for GMM/label-skew/$K = 10$, but shifted ℓ_1 mass by 1.222, above budget; the constrained difference there was −0.001746.

Table 4.1: Fixed-target comparison at successive evidence stages. Negative Δ favours constrained weights; the gain threshold is −0.005.

Evidence stage	Cells	Mean Δ	Gain slices
Adult development	54	+0.002428	0/18
Adult frozen confirmation	12	−0.000215	0/6
Bank external replication	12	−0.001332	0/4

Validation selected the copula globally. The procedure was frozen before official-test scoring, with fitting seeds 404 and 505 and the same assignments and client counts. The size-weighted primary loss averaged 0.022216; uniform weighting increased loss by 0.041139 overall, particularly under label skew. Constrained weighting changed it by −0.000215, and none of the six partition/K slices met the margin. The official test had previously been read for source validation and duplicate quarantine; the freeze preceded its first logged confirmation scoring, not its first physical access.

The external Bank Marketing replication used 41,188 records, removed post-contact duration, retained unknown categories and learned support metadata from training data. A duplicate-group-aware split held out 4,119 assessment records. Copula and Chow–Liu were transferred to age-plus-education and label-skew assignments at $K = 5$, with seeds 611, 622 and 633 and five 10,000-row draws per cell. The most favourable slice, Chow–Liu under label skew, had $\Delta = -0.004008$; copula under age-plus-education had $\Delta = +0.000441$. The split was not chronological and repeated contacts cannot be identified as independent people. The transferred 0.005 margin is an operational criterion, whose substantive meaning need not be identical across datasets.

A related question is whether regrouping client distributions into overlapping subsets

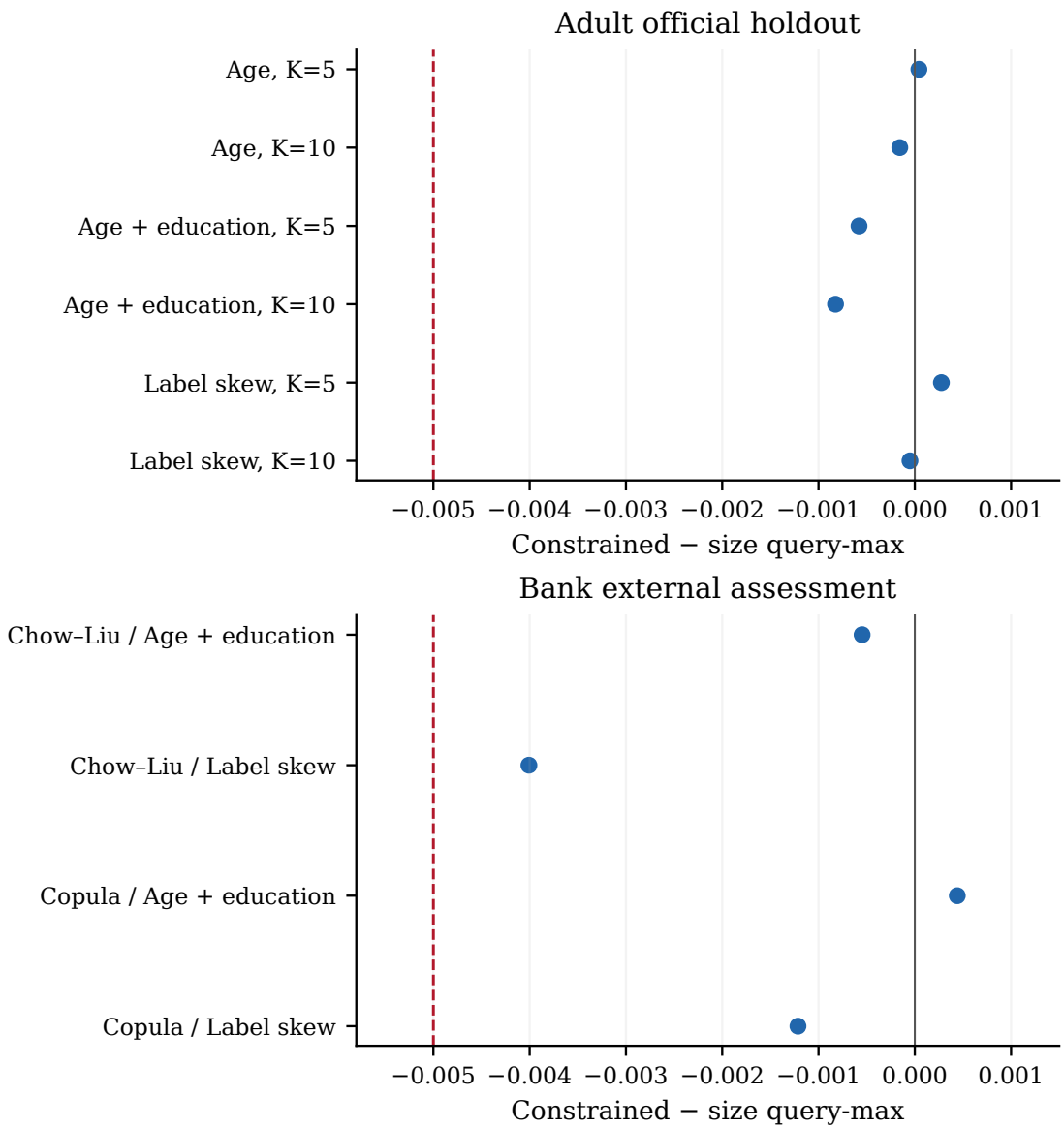


Figure 4.1: Frozen Adult and external Bank slice comparisons. Each point summarizes paired cell means; negative differences favour the constrained rule. The dashed line is the practical-gain threshold. Adult slices average two fitting seeds and Bank slices three; each cell averages five draws.

creates a richer mixture. It does not when the distributions themselves are fixed: if subset s mixes clients with weights a_{si} and subsets are mixed with weights b_s, the resulting client mass is $\sum_s b_s a_{si}$. This is an ordinary client mixture. A saved numerical check agrees with the identity. Training new estimators within subsets could change the components and was not tested. Size weighting remains the fixed-target fallback here; equivalence is not established.

4.2 Study 2: Pilot accuracy and synthetic table size

A finite synthetic pilot supplies the constrained linear program (LP). To examine pilot error and final-table noise, six fixed development checkpoints were reused: Chow–Liu age/$K = 5$ and GMM label-skew/$K = 10$, each at three seeds. Pilot sizes of 500, 1,500, 5,000 and 15,000 rows per client were each repeated five times. A separate 30,000-row-per-client synthetic matrix provided a finite reference, and one fresh 10,000-row assessment table was generated per fit. All constrained LP certificates succeeded.

At the largest pilot, mean excess objective on the independent finite reference matrix was below 0.001 for both families (Table 4.2). Yet their fresh-table differences from size were +0.000262 and +0.005894. The certificate applies to the supplied finite matrix, target and constraints, not the unknown expectation matrix, all queries or a nonlinear pooling optimum. The reference is not a population oracle.

Table 4.2: Pilot and output-size sensitivity on six reused development checkpoints. Ranges are five-draw loss ranges, averaged across three seeds.

Measure	Chow–Liu age/$K = 5$	GMM label/$K = 10$
15,000-row pilot: reference LP excess	0.000832	0.000878
15,000-row pilot: fresh Δ	+0.000262	+0.005894
Size loss range, $m = 2{,}500$	0.009476	0.015467
Size loss range, $m = 40{,}000$	0.002542	0.003472
Δ at $m = 40{,}000$	+0.006713	+0.001492

The second part fixed checkpoints and weights and varied final table size m among 2,500, 10,000 and 40,000, with five draws each. The five-draw range in size-weighted loss narrowed by roughly 73% and 78% for the two families, respectively. Error nevertheless remained at the largest size, and no family/size mean achieved the practical gain. Figure 4.2 shows the pilot and output-size results together. Some output sizes may share random-stream prefixes; the three seeds also reuse Adult people. The persistent error is an observed floor over these checkpoints and sizes, not a population lower bound.

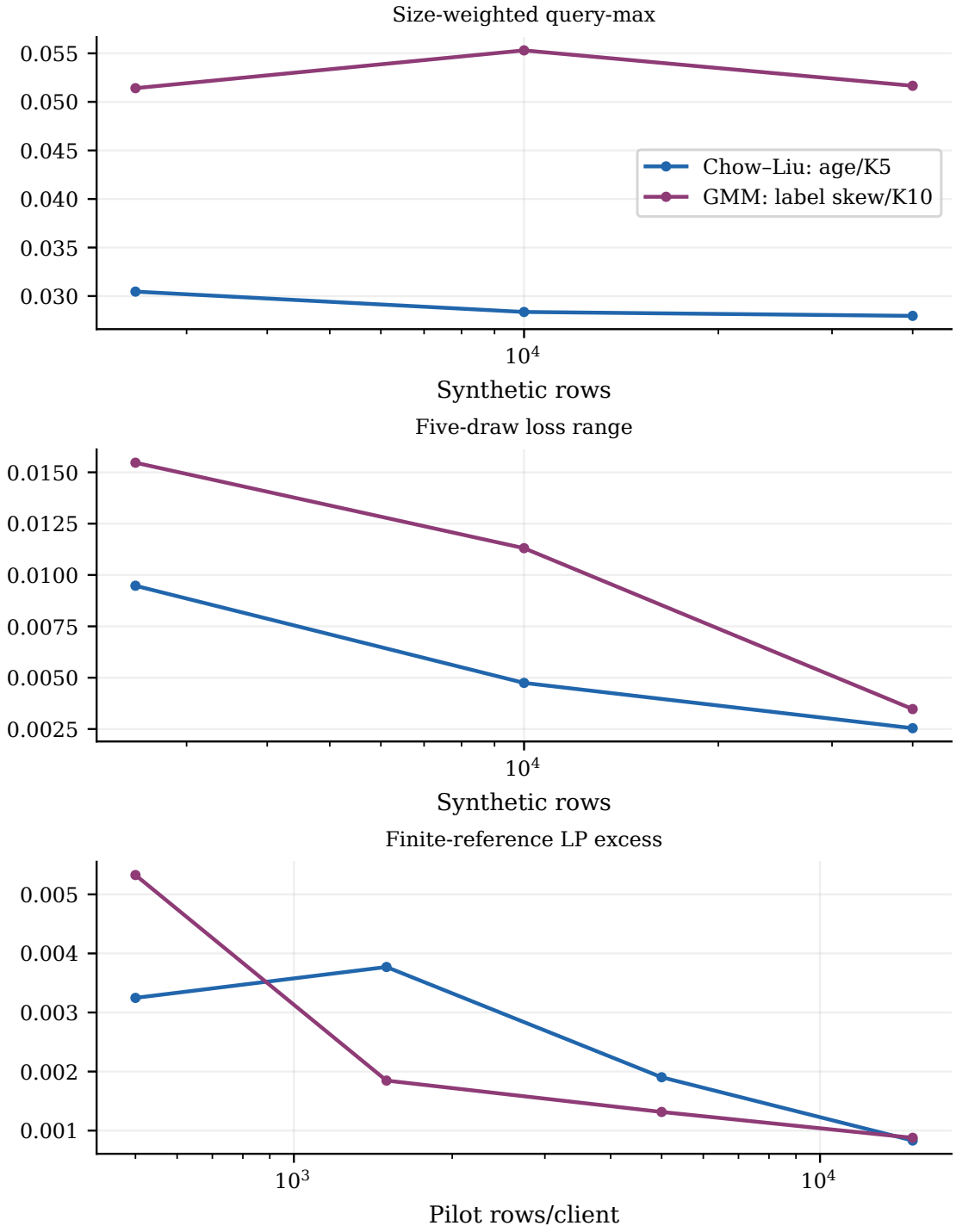


Figure 4.2: Sensitivity to output and pilot sizes in six development checkpoints. From top to bottom: size-weighted query loss, the within-cell five-draw range, and finite-reference LP excess. Lines connect tested settings rather than fitted asymptotic laws.

4.3 Study 3: Conditional generation and transfer

Conditional validity requires both filter compliance and fidelity of remaining attributes. Development compared teacher rejection with separately trained, per-condition Chow–Liu students, using the same six checkpoints and both weight rules. Separate 30,000-row teacher pools trained students and supplied direct rejection; accepted counts were matched up to 2,000. For age and bachelor's-plus-age conditions, direct rejection had lower mean maximum feature-query error in each family/weight comparison. Raw student filter violations for these requests were approximately 20–24%; postfilter samples had zero recorded filter or mapped education-code violations. Postfilter validity does not describe raw acceptance.

Frozen Adult confirmation used the direct service. Each of the three common requests produced 1,000 accepted rows in all 12 cells with zero recorded filter violations, but their feature fidelity differed sharply (Table 4.3). The rare older-doctorate request abstained in every cell; in development it failed the prior teacher gate of at least 300 rows and 1% support in every route.

Table 4.3: Frozen direct conditional service: means over 12 Adult cells. Common requests each returned 1,000 accepted rows per cell.

Condition	Mean feature-query maximum	Abstentions
Age 25–45	0.083787	0/12
Bachelor's and age 30–60	0.147013	0/12
Female	0.375000	0/12
Rare older doctorate	—	12/12

For the female request, `relationship=Husband` was the worst query in all 12 confirmation cells. Its alpha-standardized frequency was zero in the empirical female reference, while generated frequency ranged from 0.340 to 0.399 (Figure 4.3). This dependency failure occurred despite filter compliance; the support gate and filter do not guarantee fidelity for other attributes.

The transfer audit examined 48 finite empirical conditional-bound rows and 5,112 supported student/direct query triangles, using 142 feature-only queries and 6,816 teacher query answers. The recorded inequalities held in every checked row. These checks are

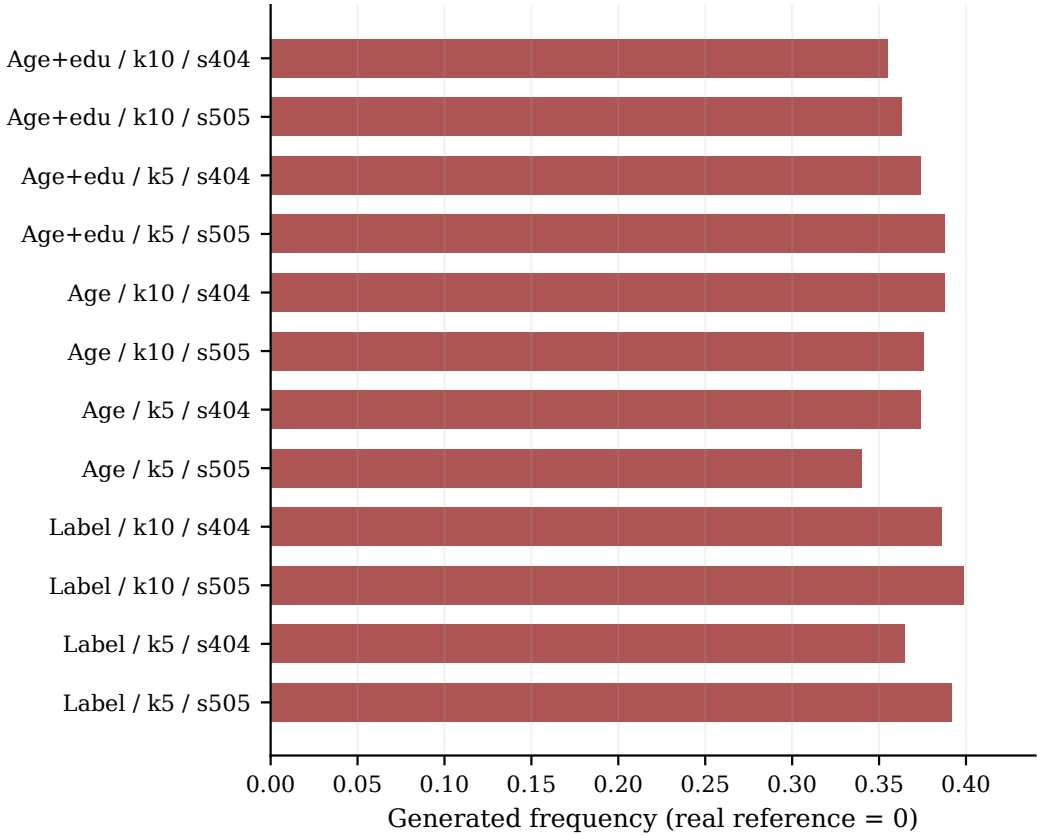


Figure 4.3: Generated frequency of `relationship=Husband` under the female condition in the 12 frozen cells. The corresponding alpha-standardized empirical reference frequency is zero in every cell; accepted rows satisfy the female filter.

finite and empirical: the ratio requires measured intersections and positive condition mass, and can be uninformative for a small denominator. They do not establish arbitrary-condition generalization, real-support guarantees or population confidence bounds.

4.4 Study 4: Downstream utility and joint-implied prediction

Frozen confirmation trained logistic and extra-trees income models on synthetic data and evaluated them on the same Adult people as the real-trained reference. Synthetic train/tune/calibration budgets were 10,000/2,000/2,000 rows, compared with 21,148/4,880/3,254 real records for the privileged train-real/test-real (TRTR) reference. Scores were alpha-standardized with source-specific calibration.

Table 4.4: Frozen Adult predictive utility at the registered, unequal training budgets; means over 12 cells. Lower Brier and log loss, and higher AUROC, are better.

Learner	Training source	Brier	AUROC	Log loss
Logistic	Synthetic, size	0.129818	0.856167	0.402970
Logistic	Synthetic, constrained	0.129254	0.857161	0.401321
Logistic	Real, TRTR	0.101987	0.904646	0.319003
Extra trees	Synthetic, size	0.133218	0.846439	0.413030
Extra trees	Synthetic, constrained	0.132705	0.846115	0.411171
Extra trees	Real, TRTR	0.105741	0.899013	0.329882

The real-trained reference performed better at these budgets (Table 4.4). Constrained weighting changed the synthetic logistic Brier score from 0.129818 to 0.129254 and AUROC from 0.856167 to 0.857161. No matched-size learning curve was run, so the gaps do not establish an intrinsic penalty at equal sample size. Group AUROC claims required at least 20 positives and 20 negatives; unsupported groups were not filled with global scores.

A second route scores both income labels under the fitted joint distribution and normalizes their masses. In the six development reference cells, size-weighted Chow–Liu gave mean Brier 0.128481 against 0.137953 for its synthetic-trained logistic comparator. GMM gave 0.126214 against 0.116265 for logistic. The Chow–Liu mass is exact for its declared discretized fitted model; the rounded and clipped GMM output score is approximate. Both labels had finite scores for all observed assessment rows, not necessarily arbitrary off-support inputs. The family comparison reverses direction.

4.5 Study 5: Query diagnostics, projection and heterogeneity

Which queries drive the maximum, and which design changes explain it? The saved development audit included 10,500 answers across 175 queries; 23 extra audit queries were inspected during development and their later values are secondary descriptions, not an independent post-hoc confirmation. In frozen confirmation, the conjunction of high-school graduation and high income was the dominant primary error in 33 of 60 size-weighted draws. This is a finite-workload diagnostic, not a bound on arbitrary queries or total variation.

The projection ablation applied the public education-to-number mapping to exactly the same raw generated rows at 12 checkpoints. Primary maxima fell substantially for both tested families (Table 4.5) and relation violations became zero. The workload includes the repaired relation, so this transformation does not establish improvement for every query. Historical unprojected and corrected results should not be pooled.

Table 4.5: Mechanism checks: projection ranges and equal-quantity heterogeneity means.

Comparison	Before / size	After / constrained
Chow–Liu projection: primary maximum	0.268–0.328	0.028–0.033
GMM projection: primary maximum	0.693–0.708	0.049–0.055
Low severity: mean primary loss	0.034804	0.033317
High severity: mean primary loss	0.034918	0.033541

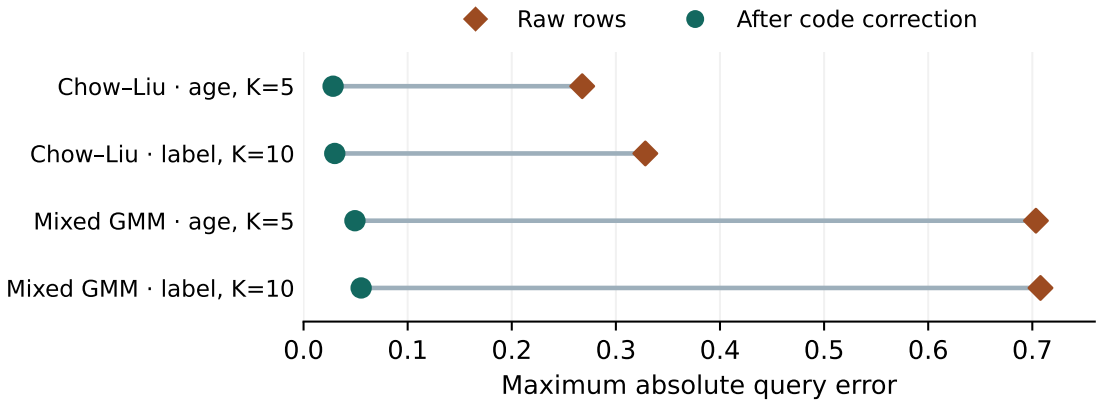


Figure 4.4: Each segment connects the error on the same raw and corrected rows, holding size weights fixed. A point is the mean over three seeds and five draws per seed. The workload includes the education-code query.

A separate access ablation held six copula checkpoints and pilots fixed while allowing 143 one-way, 151 one/two-way, or all 152 primary queries into the summary-based procedure. Mean constrained-minus-size differences were +0.000208, +0.001502 and +0.001502, respectively. The 151- and 152-query solutions coincided in all six cells because the added structural query was already zero on the compared pilots and targets. This redundancy does not imply that arbitrary higher-order information is unhelpful.

Finally, six newly fitted copula cells varied assignment severity while fixing $K = 5$, 1,000 training and 200 tuning rows per client, and equal target masses $\alpha_i = 1/5$. Three seeds were crossed with severity settings 0.25 and 0.85; five 10,000-row tables were drawn

per cell. The range of client mean training ages rose from approximately 5.1–5.8 years to 21.3–21.8 years. Nevertheless, the constrained improvement was below 0.005 at both settings, and per-seed severity contrasts changed sign. Equal quantities do not isolate every assignment or conditional-distribution change; this equal-target experiment differs from the main natural-size estimand.

4.6 Study 6: Privacy, release and communication

The main synthesis pipeline and its released summaries are nonprivate. An independent public-bin histogram provides a separate conditional privacy reference: with public membership, sizes and bins fixed, 15 released columns have replacement sensitivity 30, and fresh Laplace count noise at scale $30/\epsilon$ gives the stated pure-DP mechanism. Three overlapping fits at $\epsilon = 1, 4, 8$ compose to $\epsilon = 13$ under those assumptions; additional overlapping releases need additional accounting. The matched nonprivate histogram has approximately 0.33 primary error, indicating substantial misspecification for this workload. An attack AUC near 0.5 is an attack diagnostic, not a privacy certificate.

Table 4.6: Release and resource evidence. Privacy accounting is conditional on the stated per-fit assumptions and does not cover the main pipeline.

Release or measure	Observed result
Main pipeline	Nonprivate model and summary release
Histogram triplet	$\epsilon = 1 + 4 + 8 = 13$ under fixed public membership, sizes and bins
Nonprivate Private-PGM	Four fits; mean $\Delta = +0.002615$ versus size
Adapted FLAIM	Four fits; mean $\Delta = +0.012894$ versus nonprivate size; basic composition $(16, 4\times10^{-6})$ under per-fit assumptions
Model plus metadata	1.915–2.866 times matched raw-table bytes in three copula cells

The four nonprivate Private-PGM fits [8] used the same released unnoised summaries and produced one favourable and three adverse cell differences from size. Four adapted author-code FLAIM fits each reported $(\epsilon, \delta) = (4, 10^{-6})$. Its interactive raw-client contract and private marginals differ from the model/summary contract, and the comparator is nonprivate. These are not equal-access or equal-privacy rankings and do not protect the main release.

Communication was measured using the same pickle protocol 5 and gzip level 6 for raw tables and model-plus-metadata payloads in three copula cells. Their ratios were 2.866 for age-plus-education/$K = 5$, 1.984 for age-plus-education/$K = 10$, and 1.915 for label skew/$K = 10$. Aggregate summary traffic added 145,317, 157,267 and 160,769 bytes, respectively. There was no measured compression benefit. Unix-socket application bytes exclude WAN, TLS, authentication and retries.

Table 4.7: Application payloads in bytes for the matched copula serialization audit, seed 101. The optional summary is additional to the model-plus-metadata column.

Partition and clients	Raw table	Model + metadata	Summary
Age and education, $K = 5$	265,949	762,098	145,317
Age and education, $K = 10$	287,339	570,000	157,267
Label skew, $K = 10$	282,648	541,373	160,769

To interpret the privacy result, let D, D' differ by replacement of one record and let E be any observable release event. A mechanism M is (ϵ, δ)-differentially private if

$$\Pr[M(D) \in E] \leq e^{\epsilon} \Pr[M(D') \in E] + \delta$$

for every such pair and event [4]. A replaced record can decrease one bin and increase another in each column. Across fifteen columns this gives the histogram's sensitivity bound of thirty. This argument accounts for publishing noisy counts with fixed public bins; it does not account for separately publishing a fitted copula. Table 4.7 answers the distinct engineering question of how many serialized bytes were actually sent.

Chapter 5

Guarantees, Optimality, and Limits

The following results explain when size weights are justified and what a finite-workload comparison can establish. They are conditional mathematical statements, not a claim that the experiments verified their hypotheses or established a universal certificate.

5.1 When size weights are optimal

Theorem 5.1.1 (Exact-model size optimality)**.** *Let the population target be* $P_\alpha = \sum_i \alpha_i P_i$ *and suppose* $Q_i = P_i$ *for every client. Then* $w = \alpha$ *minimizes both* $\mathrm{KL}(P_\alpha \| Q_w)$ *and* $\max_{q \in \mathcal{Q}} |\mathbb{E}_{P_\alpha} q - \mathbb{E}_{Q_w} q|$*, with value zero. Affine independence of the component measures is sufficient for a unique KL minimizer; affine independence of their query-answer vectors is sufficient for a unique query-loss minimizer. These conditions are necessary for global injectivity, but a boundary point may have unique weights without global injectivity.*

Proof. The assumptions give $Q_\alpha = P_\alpha$. Both discrepancies are nonnegative and vanish at equality. Any other zero-KL solution equals the target measure, so affine independence identifies its weights. For query loss the same argument applies to the vectors of query expectations. Conversely, a nonzero zero-sum kernel direction can be scaled around an interior simplex point to give two equal mixtures or answer vectors; this does not preclude uniqueness at a particular boundary point. □

On a finite discrete support, expected negative log likelihood under P_α is its entropy plus $\mathrm{KL}(P_\alpha \| Q_w)$, when well defined. Separately, a multinomial likelihood for indepen-

dently sampled client labels is maximized by observed label proportions; client counts fixed by design do not establish that sampling model. Neither argument proves that α maximizes the *finite observed-data mixture likelihood*. For example, exact Bernoulli components with probabilities 0.2 and 0.8 and $\alpha = (1/2, 1/2)$ yield likelihood $0.2w_1 + 0.8w_2$ for one observed success, maximized at $(0, 1)$.

Remark 5.1.2 (Misspecification can favour another feasible weight). Let $P_1 =$ Bernoulli(0.2), $P_2 =$ Bernoulli(0.8), $\alpha = (0.5, 0.5)$, $Q_1 =$ Bernoulli(0.3) and $Q_2 = P_2$. The target success probability is 0.5. Size weights give 0.55, while $w = (0.6, 0.4)$ gives 0.5 exactly. This alternative obeys both constraints in (2.4): its ℓ_1 distance is 0.2 and both floors hold. For $q(z) = z$, losses are 0.05 and zero, respectively; the corresponding target-to-mixture KLs are approximately 0.005025 and zero. Small local model error therefore does not imply exact size optimality.

Proposition 5.1.3 (Approximate size optimality)**.** *Let* $e_i = \|H_{\cdot i} - B_{\cdot i}\|_\infty$, $e_\alpha = \sum_i \alpha_i e_i$, *and* $e_{\max} = \max_i e_i$. *If a closed feasible set* $\mathcal{W} \subseteq \Delta_K$ *contains* α *and* w^* *minimizes* L_R *over it, then*

$$0 \le L_R(\alpha) - L_R(w^*) \le e_\alpha. \tag{5.1}$$

If additionally $\|Bv\|_\infty \ge c\|v\|_1$ *for all* v *with* $\mathbf{1}^\top v = 0$, *for some* $c > 0$, *then* $\|w^* - \alpha\|_1 \le (e_\alpha + e_{\max})/c$.

Proof. Since $b_R = B\alpha$, $L_R(\alpha) = \|(H - B)\alpha\|_\infty \le e_\alpha$. Optimality and nonnegativity give the first bound. Moreover $L_R(w) \ge \|B(w - \alpha)\|_\infty - \|(H - B)w\|_\infty \ge c\|w - \alpha\|_1 - e_{\max}$. Apply this at w^* and use $L_R(w^*) \le e_\alpha$. □

The localization condition is strong: it fails when the reference workload cannot distinguish some zero-sum changes in client weights.

Proposition 5.1.4 (A sufficient dual certificate)**.** *Let* $r = H\alpha - b_R$ *and* $t = \|r\|_\infty > 0$. *If a vector* s *is supported on the active coordinates* $\{\ell : |r_\ell| = t\}$, *has* $s_\ell = \lambda_\ell \operatorname{sign}(r_\ell)$ *for nonnegative* λ_ℓ *summing to one, and satisfies* $H^\top s = \nu\mathbf{1}$ *for some scalar* ν, *then* α *globally minimizes* L_R *over the simplex. If* $t = 0$, *it is already optimal.*

Proof. Here $\|s\|_1 = 1$ and $s^\top r = t$. For any simplex w, norm duality gives $\|Hw - b_R\|_\infty \ge s^\top(Hw - b_R) = t + \nu\mathbf{1}^\top(w - \alpha) = t$. Size weights attain t. □

This is a sufficient certificate, including at simplex boundaries, rather than a necessary one in general. It has not been checked for the campaign's exact answer matrix.

5.2 Geometry, representation, and estimation

The achievable model-answer set is $H\Delta_K = \operatorname{conv}\{H_{\cdot i}\}$, hence $\inf_{w\in\Delta_K} L_R(w) = \operatorname{dist}_\infty(b_R, H\Delta_K)$. The constrained attainable set is $H\mathcal{W}$. Reweighting fixed models cannot reach a target outside that set. Different weights can nevertheless have identical answers: global weight identification requires $\ker H \cap \{v : \mathbf{1}^\top v = 0\} = \{0\}$. Mixing mixtures of the same components only regroups their coefficients and does not enlarge the convex hull.

Proposition 5.2.1 (Sharp representation bound)**.** *For $H \in [0,1]^{L\times K}$ define $D_H = \max_{i,j} \|H_{\cdot i} - H_{\cdot j}\|_\infty$. Then*

$$\|H(w-\alpha)\|_\infty \le \frac{D_H}{2}\|w-\alpha\|_1 \le \frac{1}{2}\|w-\alpha\|_1. \tag{5.2}$$

The constant $1/2$ is sharp.

Proof. Write $v = w-\alpha$. Its positive and negative parts have equal mass $s = \|v\|_1/2$. After normalization each is a probability vector, so for each row h, $|hv| \le s(\max_i h_i - \min_i h_i)$. Maximizing proves the result. Equality in the second bound occurs for $h = (0,1)$, $\alpha = (1,0)$ and $w = (1-t, t)$. □

The 0.25 budget therefore limits the change in any bounded model query relative to Q_α to 0.125; it does not bound error relative to real data. In fact, the exact identity $b_S - B\alpha = (b_S - Hw) + (H-B)\alpha + H(w-\alpha)$ and (5.2) give

$$\begin{aligned}\|b_S - B\alpha\|_\infty \le \|b_S - Hw\|_\infty + \sum_i \alpha_i \|H_{\cdot i} - B_{\cdot i}\|_\infty \\ + \frac{D_H}{2}\|w-\alpha\|_1.\end{aligned} \tag{5.3}$$

This is an upper bound, not a causal attribution: terms can cancel. An additional aggregation or compression discrepancy must be added when its output differs from Q_w.

Theorem 5.2.2 (Pilot oracle bound)**.** *Suppose* $\max_i \|\widehat{H}_{\cdot i} - H_{\cdot i}\|_\infty \le r_H$ *and* $\|\widehat{b} - b_R\|_\infty \le r_b$*. Set* $r = r_H + r_b$ *and* $\widehat{L}(w) = \|\widehat{H}w - \widehat{b}\|_\infty$*. If* $\widehat{w}$ *minimizes* (2.5) *to tolerance* ζ *and* w^* *minimizes* L_R *over* $\mathcal{W}$*, then*

$$\begin{aligned} |\widehat{L}(w) - L_R(w)| &\le r \qquad\qquad (w \in \Delta_K), \\ L_R(\widehat{w}) - L_R(w^*) &\le 2r + \frac{\lambda}{2}\|w^* - \alpha\|_2^2 + \zeta. \end{aligned} \tag{5.4}$$

Proof. For any simplex weight, $\|(\widehat{H} - H)w\|_\infty \le r_H$; the reverse triangle inequality gives the uniform bound. Compare the approximate regularized minimum with w^*, use the uniform bound at both weights, and discard the nonnegative penalty at $\widehat{w}$. □

The uniform event permits data-adaptive weight selection from the same pilot, but does not cover arbitrary adaptive discovery of queries. For fixed queries and p_i independent pilot draws per client, a simultaneous Hoeffding radius is $r_H = \max_i \sqrt{\log(2KL/\gamma_H)/(2p_i)}$ with probability at least $1 - \gamma_H$. A summary radius r_b requires its own justification; exact calibration summaries do not make calibration and held-out reference identical. Combine (5.4) with (2.3) for finite-table error, under their respective conditions.

5.3 Conditional transfer and practical scope

Proposition 5.3.1 (Conditional query transfer)**.** *For fixed real and synthetic measures* R, S*, let* $r = R(C) > 0$*,* $s = S(C) > 0$*,* $e_0 = |s - r|$*, and* $e_C = \max_{q \in \mathcal{Q}_C} |\mathbb{E}_S[\mathbf{1}_C q] - \mathbb{E}_R[\mathbf{1}_C q]|$*, with* $q \in [0, 1]$*. Then*

$$\max_{q \in \mathcal{Q}_C} |\mathbb{E}_S[q \mid C] - \mathbb{E}_R[q \mid C]| \le \min\left\{1, \frac{e_C + e_0}{r}\right\}. \tag{5.5}$$

Proof. For $a = \mathbb{E}_R[\mathbf{1}_C q]$ and $c = \mathbb{E}_S[\mathbf{1}_C q]$, add and subtract c/r in $|c/s - a/r|$. The result is at most $|c - a|/r + c|r - s|/(rs) \le (e_C + e_0)/r$, since $c \le s$. Conditional differences are at most one. □

Rare reference conditions amplify errors; if $s = 0$, the synthetic conditional is undefined and the method must abstain. Matching one-way marginals alone does not

establish conditional fidelity: equal mass on $(0,0),(1,1)$ and on $(0,1),(1,0)$ gives identical one-way means but opposite $Y \mid X$ relationships. Predictive transfer similarly needs a loss-class discrepancy and a generalization bound, not merely the observed workload maximum.

Further reviewed proofs appear in `theory/proofs.tex`, with qualifications in `theory/PROOF_REVIEW.md`. They cover sampling, projection, finite support, conditional generation, prediction, aggregation and privacy. In particular, the post-experiment fallback needs fresh independent candidate and baseline tables and was not the frozen selection method. Privacy composition requires valid guarantees for every released mechanism: parallel composition further assumes fixed disjoint membership, client-local mechanisms and independent client randomization conditional on fixed data; overlapping access requires appropriate sequential composition. Postprocessing preserves an established guarantee, but the main release is not thereby certified private.

These conditional results do not establish universal superiority of the reported method. Population discrepancies and local model errors in the bounds are not automatically observed in the finite-query experiment.

Chapter 6

Conclusion

For the target studied here, the practical decision is to keep the projected copula and sample its clients in proportion to their training counts. The constrained rule sometimes gave a smaller error. Its reductions on frozen Adult and on Bank, averaged over their respective slices, did not reach the improvement specified before scoring. That observation supports a choice between the two recorded procedures; it does not settle weighting for a different set of models, clients or queries.

The most informative contrast is between changing *which generator supplies a row* and changing *what that generator can emit*. The education-code correction makes a specified inconsistency impossible for mapped categories. Reweighting alone can only move between the query vectors already offered by the generators. At the same time, satisfying a requested condition does not establish the quality of the records within that condition: the female-conditional relationship audit exposed a substantial mismatch even on accepted rows. For a data user, these distinctions matter more than a small improvement in a global summary.

A follow-up could fix new interaction queries, learn a local correction, and test it on new held-out records. A private comparison would also require matched access and privacy budgets. The Adult test used here can no longer serve as an untouched selection set.

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Appendix A

Guide to the evidence

The code, saved results and complete mathematical supplement are available in the project repository: S-G-mathematics/btp-federated-synthetic-data. The version used to prepare the report is commit `f1624441be85`. The repository retains twelve technical work packages; the six studies in this report group them by scientific question. The grouping changes no runs, denominators or results.

Table A.1: Study-to-evidence crosswalk. WP numbers refer to the permanent entries in the repository's `experiments/` directory.

Study	WP	Evidence to inspect
1	1, 3	Corrected weighting slices, validation selection, frozen Adult and external Bank tables; subset-composition identity.
2	2, 4	Pilot-size sweeps, LP gaps and independent evaluation; draw-size ranges.
3	5, 11	Teacher and student requests, rejection and abstention, conditional transfer inequalities.
4	6, 7	Real- and synthetic-trained utility, calibration and joint-implied scores.
5	8, 9, 10	Dominant queries, paired projection, summary-access contrasts, fixed-quantity severity.
6	12	Mechanism-specific privacy ledger, adapted references, serialized payload sizes.

The proof supplement gives all 27 statements, assumptions and counterexamples; its later certificates are not inputs to the historical selection decision. The reproduction guide distinguishes rebuilding tables from saved evidence from rerunning model fits. Source data and frozen selections have independent checksums. The principal figures are based on the saved numerical results, not digitized images.

The wider development comparison

Before freezing one challenger, the development benchmark compared thirteen size, score, calibration and pooling rules. Their means clarify why representation and access matter: moment matching reduced the average loss by 0.000764 but moved the client weights by 0.590 in ℓ_1 on average. A small aggregate improvement can therefore require a substantial change in who contributes the rows. Table A.2 retains all candidates, including methods with scores larger than the size baseline.

Table A.2: Thirteen source candidate methods over the 54 corrected Adult core cells; descriptive assessment means.

Method	Access class	Query-max	Δ vs. size	Mean $\|w - \alpha\|_1$
Size	Models/counts	0.035633	+0.000000	0.000
Uniform	Models/counts	0.049719	+0.014086	0.391
Moment least squares	Models/summaries	0.034868	-0.000764	0.590
Entropy calibration	Models/summaries	0.035162	-0.000471	0.152
Residual GLS	Models/summaries	0.035031	-0.000601	0.559
Kernel mean matching	Models/summaries	0.035868	+0.000235	0.738
Unconstrained query minimax	Models/summaries	0.039671	+0.004038	0.969
Robust query calibration	Models/summaries	0.037909	+0.002277	0.806
Predictive-score softmax	Privileged scores	0.134920	+0.099288	1.588
Predictive evidence	Privileged scores	0.134920	+0.099288	1.588
Stacking	Privileged scores	0.049619	+0.013986	0.837
Finite geometric pool	Models/summaries	0.298326	+0.262693	1.441
Finite entropic barycenter	Models/summaries	0.103013	+0.067381	0.414

Each method/cell averages five draws. Three separate independent-histogram control cells are excluded. The representation-constrained challenger is a separately audited variant (Table 4.1). For nonlinear pools the last column refers to their recorded input weights, not an equivalent output client-mixture mass. Privileged scores, finite nonlinear supports, and different released summaries prevent an equal-information ranking. These are implementation outcomes, not optimal values for each literature family.

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